

PROJECT REPORT

Amazon Customers Sentiment Analysis

An interactive Power BI dashboard on 49,981 customer reviews

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Abstract

This project analyses 49,981 Amazon customer reviews to find out how satisfied customers are and what drives their opinions. Each review was scored with the VADER sentiment analyser and classified as Positive, Negative or Neutral. The results were presented in a three-page interactive Power BI dashboard (Overview, Trends and Insights).

The analysis shows that **87.27%** of reviews are positive, **10.54%** are negative and **2.19%** are neutral, with an average star rating of **4.19**. Sentiment stays stable over the years even as the number of reviews grows. Taste, flavour and packaging are the most discussed topics, and the report ends with practical recommendations based on these findings.

49,981 Total Reviews	4.19 Avg Rating	0.64 Avg Sentiment	10.54% Negative	87.27% Positive
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1. Introduction and Objectives

Online reviews are one of the richest sources of customer feedback, but reading thousands of them by hand is not practical. Sentiment analysis solves this by turning free-text reviews into measurable signals that a business can track and act on.

This project, "Amazon Customers Sentiment Analysis", applies sentiment analysis to a large set of Amazon reviews and presents the outcome as an interactive dashboard. The objectives were:

- To measure overall customer satisfaction using star ratings and VADER sentiment scores.

- To track how ratings and sentiment change over time (2006 to 2012).
- To identify the words and topics that customers use most in positive and negative reviews.
- To compare star ratings with text sentiment and find where they disagree.
- To give clear, data-backed recommendations to improve customer experience.

2. Tools and Technologies

Tool	Purpose in this project
Microsoft Power BI Desktop	Building the three-page interactive dashboard, slicers and visuals
DAX	Calculated measures such as Total Reviews, Positive %, Negative % and Avg Sentiment Score
VADER	Scoring each review's text and labelling it Positive, Negative or Neutral
Power Query	Cleaning and transforming the data before analysis

3. Dataset and Methodology

The dataset contains 49,981 Amazon customer reviews written between 2006 and 2012. The main fields used are listed below.

Field	Description
ProductId	Identifier of the reviewed product
Score	Star rating from 1 to 5
Summary and Text	Review headline and full review text
Review_Length	Length of each review
Sentiment_VADER	VADER label: Positive, Negative or Neutral
Sentiment_Rating	Star rating grouped into Positive, Neutral or Negative
Year, Month, YearMonth	Time fields used for trend analysis
Top_words	Separate table of frequent words with their sentiment and count

The work followed these steps:

1. Load the review data into Power BI and clean it (remove unusable rows, create Year, Month and YearMonth fields).
2. Apply VADER to the review text to get a sentiment score and a Positive, Negative or Neutral label.
3. Group star ratings into Positive, Neutral and Negative so they can be compared with VADER results.
4. Extract the most frequent words in positive and negative reviews into the Top_words table.

5. Create DAX measures for Total Reviews, Avg Rating, Avg Sentiment Score, Positive % and Negative %.
6. Design three dashboard pages with slicers for Year, Sentiment_VADER and Score.

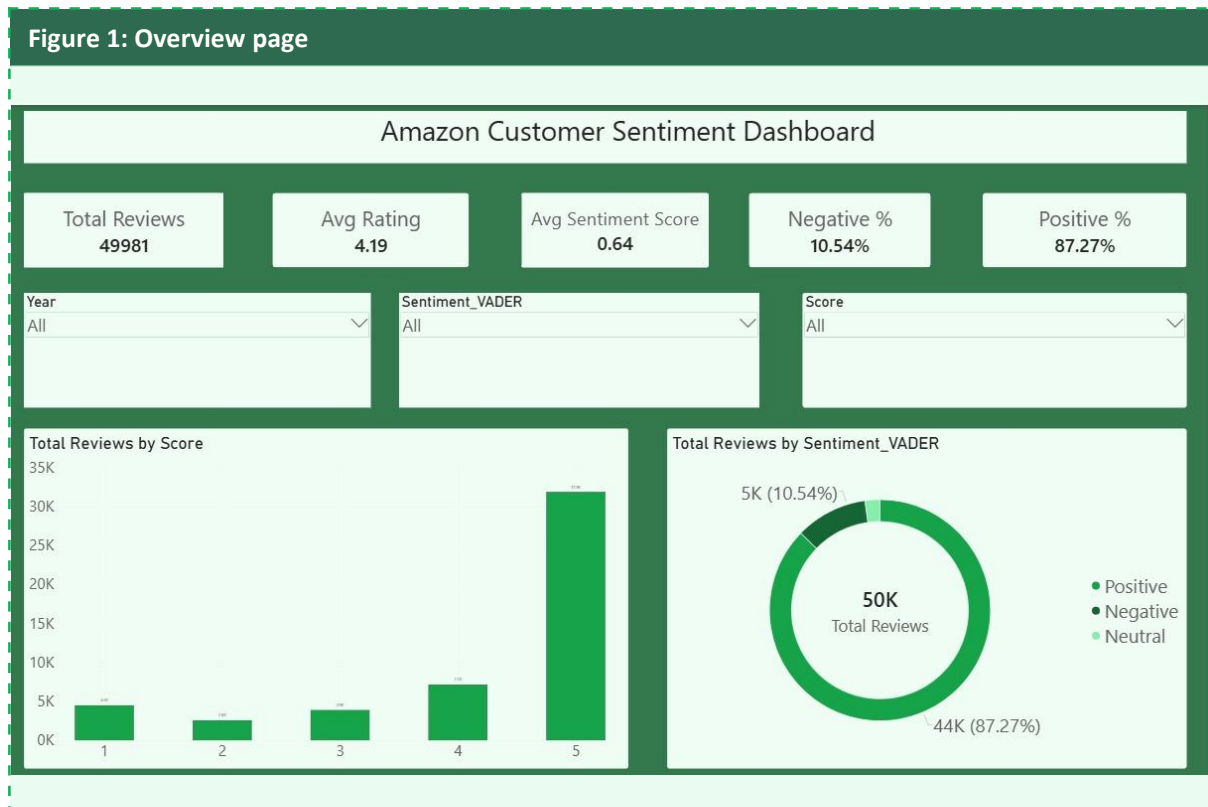
4. Dashboard Overview

The dashboard has three pages. Each page is described below, with space left for its screenshot.

4.1 Overview page

The Overview page gives the headline numbers at a glance: 49,981 total reviews, an average rating of 4.19, an average sentiment score of 0.64, 10.54% negative reviews and 87.27% positive reviews. Slicers for Year, Sentiment_VADER and Score let the reader filter every visual. A column chart shows total reviews by score, and a donut chart shows the split between Positive, Negative and Neutral sentiment.

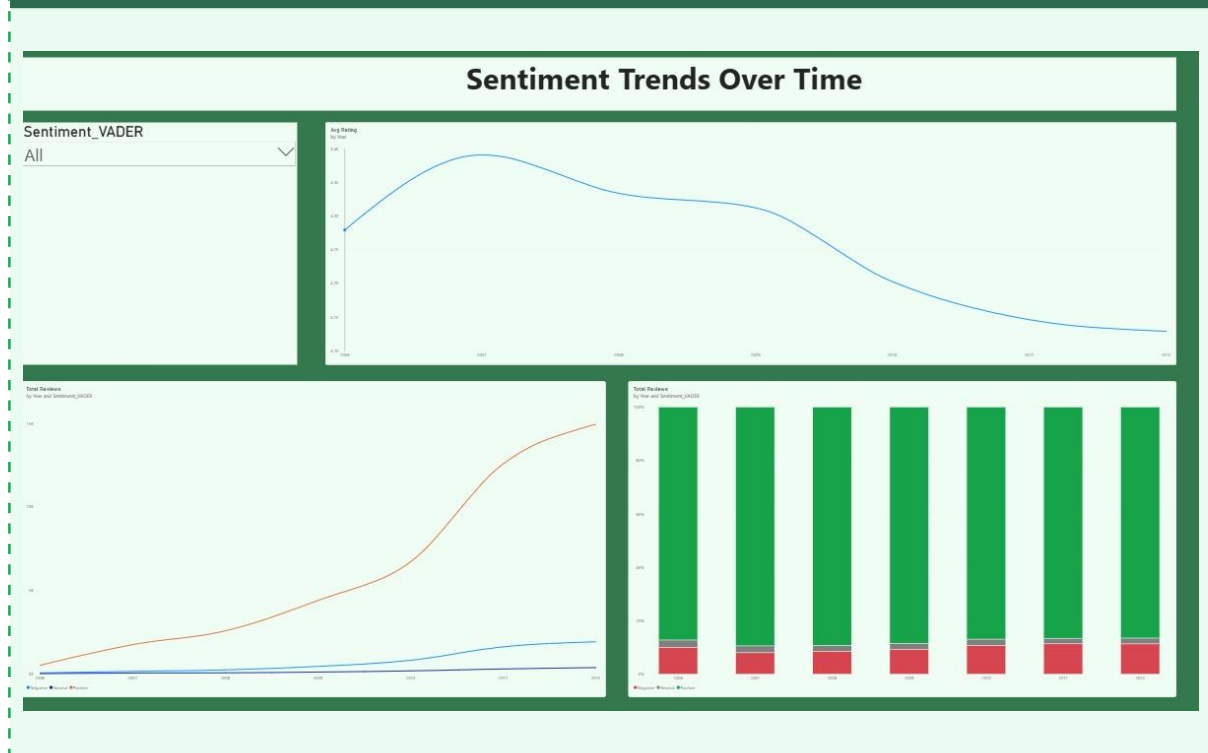
Figure 1: Overview page



4.2 Trends page

The Trends page shows how customer opinion changes over time. It contains a line chart of average rating by year, a line chart of total reviews by year and sentiment, and a 100% stacked column chart showing the sentiment share for each year. A Sentiment_VADER slicer filters the page.

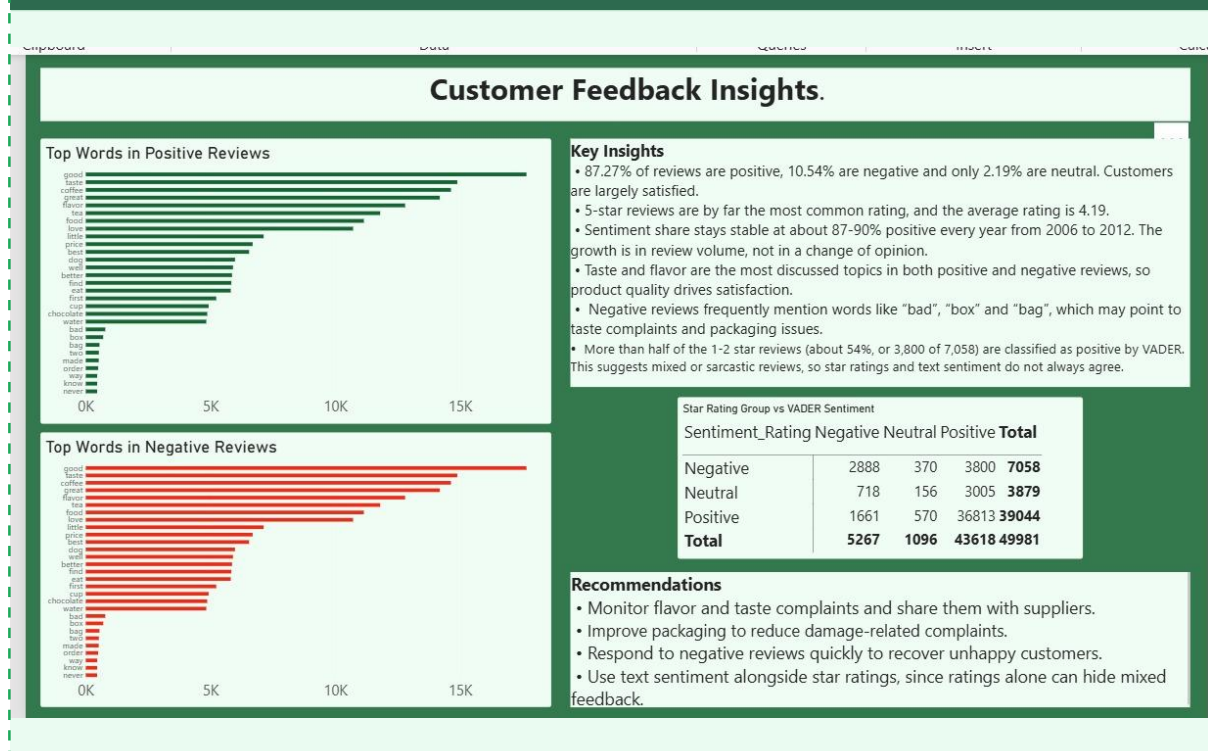
Figure 2: Trends page



4.3 Insights page

The Insights page explains the reasons behind the numbers. Two bar charts list the top words in positive and negative reviews. A matrix compares star-rating groups with VADER sentiment, and text boxes summarise the key insights and recommendations.

Figure 3: Insights page



5. Key Findings

Customers are largely satisfied: **87.27%** of reviews are positive, **10.54%** are negative and only **2.19%** are neutral.

- **Ratings are high.** 5-star reviews are by far the most common, and the average rating is 4.19.
- **Sentiment is stable over time.** About 87% to 90% of reviews are positive every year from 2006 to 2012. The number of reviews grows sharply, but opinion does not change.
- **Product quality drives satisfaction.** Taste and flavour are the most discussed topics in both positive and negative reviews. Common positive words include good, taste, coffee, great and flavor.
- **Complaints point to packaging.** Negative reviews often mention words such as "bad", "box" and "bag", which may indicate taste complaints and packaging problems.
- **Ratings and text can disagree.** About 54% of 1-2 star reviews (3,800 of 7,058) are classified as positive by VADER, which suggests mixed or sarcastic reviews.

Table 1: Star-rating group vs VADER sentiment

Star-rating group	VADER Negative	VADER Neutral	VADER Positive	Total
Negative	2,888	370	3,800	7,058
Neutral	718	156	3,005	3,879
Positive	1,661	570	36,813	39,044
Total	5,267	1,096	43,618	49,981

6. Recommendations, Limitations and Conclusion

Recommendations

- Monitor flavour and taste complaints and share them with suppliers.
- Improve packaging to reduce damage-related complaints.
- Respond to negative reviews quickly to recover unhappy customers.
- Use text sentiment alongside star ratings, since ratings alone can hide mixed feedback.

Limitations

- VADER is a lexicon-based tool, so it can misread sarcasm and mixed reviews.
- The data covers only 2006 to 2012, so results may not reflect current customer behaviour.
- Word frequency shows what customers talk about, but not always why they feel that way.

Conclusion

The project shows that Amazon customers in this dataset are overwhelmingly satisfied, and that this satisfaction has stayed steady as review volume grew. Combining star ratings with VADER sentiment

gives a fuller picture than ratings alone, and the word analysis points to taste and packaging as the areas to watch. The Power BI dashboard turns about 50,000 reviews into a clear, interactive view that supports quick decisions.

Future Scope

- Use machine-learning classifiers to improve sentiment accuracy on sarcastic reviews.
- Add topic modelling to group complaints automatically.
- Update the dashboard with newer review data.